III. LOGICA I DEMOSTRACIONS

III.1. Operacions logiques

$P \wedge Q$ = P i Q $P \vee Q$ = P o Q

$P \Rightarrow Q$ = P implica Q = si P aleshores Q $P \Leftrightarrow Q$ = P equivalent a Q = P si i només si Q

taules de veritat

$\wedge$	0	1
0	0	0
1	0	1

$\vee$	0	1
0	0	1
1	1	1

	$\neg$
0	1
1	0

$P \Rightarrow Q$	0	1 ← Q
0	1	1
P ↑ 1	0	1

$\Leftrightarrow$	0	1
0	1	0
1	0	1

prop. $\vee$ i $\wedge$ son comutatius, associatius, distributius.

1. $P \vee Q = Q \vee P$ — idem amb $\wedge$
2. $P \vee (Q \vee R) = (P \vee Q) \vee R$ — idem amb $\wedge$
3. $P \vee (Q \wedge R) = (P \vee Q) \wedge (P \vee R)$ — idem $\wedge \leftrightarrow \vee$

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prop.
1. $\neg(P \vee Q) = (\neg P) \wedge (\neg Q)$ [llei de Morgan]
2. $\neg(P \wedge Q) = (\neg P) \vee (\neg Q)$
3. $\neg(\neg P) = P$

prop.
1. $P \Leftrightarrow Q = (P \Rightarrow Q) \wedge (Q \Rightarrow P)$
2. $P \Rightarrow Q = (\neg P) \vee Q$
3. $P \Rightarrow Q = (\neg Q) \Rightarrow (\neg P)$ contraposat
4. $(\neg P) \Rightarrow 0 = P$ absurde

III.2. Quantificadors:

$\forall a \in A$ = per tot $a \in A$

$\exists a \in A$ = existeix $a \in A$

$\exists! a \in A$ = existeix un únic $a \in A$

prop.
1. $\neg \forall a \in A\ P(a) \Leftrightarrow \exists a \in A\ \neg P(a)$
2. $\neg \exists a \in A\ P(a) \Leftrightarrow \forall a \in A\ \neg P(a)$

prop.
1. $\forall a \in A\ (P(a) \wedge Q(a)) \Leftrightarrow (\forall a \in A\ P(a)) \wedge (\forall a \in A\ Q(a))$
2. $\exists a \in A\ (P(a) \vee Q(a)) \Leftrightarrow (\exists a \in A\ P(a)) \vee (\exists a \in A\ Q(a))$

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exe. Son certes les asercions següentes?

$\forall a \in A (P(a) \vee Q(a))$

$\Leftrightarrow (\forall a \in A\ P(a)) \vee (\forall a \in A\ Q(a))$

$\exists a \in A (P(a) \wedge Q(a))$

$\Leftrightarrow (\exists a \in A\ P(a)) \wedge (\exists a \in A\ Q(a))$

["cualquier numero es par o impar"
"cualquier numero es par" o
"cualquier numero es impar"]

III.3. Raonaments

* directa

exm x, y parell $\Rightarrow x+y$ parell

pr x parell $\Rightarrow \exists x'\ x = 2x'$
y parell $\Rightarrow \exists y'\ y = 2y'$
$\Rightarrow x+y = 2x'+2y' = 2(x'+y')$
$\Rightarrow x+y$ parell □

* analysi de casos

exm $\exists x, y$ irrationals tal que x^y racional

pr. analysis de casos:

* $\sqrt{2}^{\sqrt{2}} \in \mathbb{Q} \Rightarrow x = \sqrt{2}$ $y = \sqrt{2}$ convienen
* $\sqrt{2}^{\sqrt{2}} \notin \mathbb{Q} \Rightarrow x = \sqrt{2}^{\sqrt{2}}$ $y = \sqrt{2}$ convienen
perque $(\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^{\sqrt{2}\cdot\sqrt{2}} = \sqrt{2}^2 = 2$ □

* contraposat $(A \Rightarrow B) \Leftrightarrow (\neg B \Rightarrow \neg A)$

exm x^2 parell $\Rightarrow x$ parell

pr x imparell $\Rightarrow \exists x'\ x = 2x'+1$
$\Rightarrow x^2 = (2x'+1)^2 = 2(2x'^2+2x')+1$
$\Rightarrow x^2$ imparell □

* absurde / contradicio $(\neg A \Rightarrow \text{Falsa})$

exm $\sqrt{2}$ es irrational

pr Supos $\sqrt{2} = p/q$ amb $p, q \in \mathbb{Z}$ $q \neq 0$ $\text{mcd}(p,q) = 1$
$\Rightarrow 2 = \sqrt{2}^2 = p^2/q^2 \Rightarrow 2q^2 = p^2 \Rightarrow 2 | p^2 \Rightarrow 2 | p$
$\Rightarrow 4 | p^2 = 2q^2 \Rightarrow 2 | q^2 \Rightarrow 2 | q$
contradicio amb $\text{mcd}(p,q) = 1$ □

exm ∞ d'enters primers
Conjunt de conjunts no existeix

* recurrencies / induccio (initialisacio transmissio).

exm. $\sum_{i=1}^{n} i = n(n+1)/2$

pr $n = 1$ ok
$\sum_{i=1}^{n+1} i = \sum_{i=1}^{n} i + n+1 = n(n+1)/2 + n+1$
$= (n+2)(n+1)/2$ □

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